

Microphonics



Duncanamps

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Heres a question that came up on alt.guitar.amps about microphonics in tubes. Also clipped to the end is a follow up by Kevin Deal of Upscale Audio.

On Mon, 04 May 1998 09:44:57 -0800, czempel@ns.net (Chip Zempel) wrote:

>I was reading some of the amp reviews at Harmony Central, and saw several
>mentions of people having trouble with tubes that were (doing? exhibiting?
>experiencing? picking up?) microphonics. What is this? What what causes it
>and what are the symptoms? Is it fixed by replacing the tube?

Chip,

The tube is made up of a number of metal parts assembled together inside the glass envelope. If shocked mechanically, these metal parts can move, and in doing so will alter their capacitance and position such that a variation in voltage is produced.

Much the same way as a microphone. Mechanical impulses are turned into voltage. Trouble is, some tubes don't do it very much, and some do it a lot! The ones that do it a lot fit the description of "microphonic" you saw at Harmony Central. In extreme cases, you can get feedback from the amp.

If the tubes are reasonably easy to access on your amp, you could try tapping them (gently) with a pencil, and listen to them act as a microphone!

ps: not all the tubes will make the same amount of noise, a loud one may not be the most microphonic, it may just be placed where in a high gain stage where it is easier to pick up the dink-dink from the glass...

Best Regards,
Duncan

At 08:40 PM 5/5/98 +0100, Kevin wrote:

Great explanation. I just want to add a point to the original poster since on occasion guys get freaked when any tube makes a noise when tapped on. In some positions any tube you put in will make a noise if you tap it. It's normal. True microphony will rear it's ugly head and make itself known with the most obnoxious distortion you can imagine. A smidge of microphony can (at least in the hi-fi world) add a sense of "air". That's why some guys buy tube dampers and find it sounds better without them. Sometimes.

Kevin Deal Precision selected fine vintage audio tubes

Upscale Audio Authorized Sonic Frontiers, Anthem, Golden Tube

2504 Spring Terrace Audible Illusions, Presence, Margules/Magenta

Upland, CA 91784 Basis, Benz, B&K, Meadowlark, PSB, Energy

Voice(909) 931-9686 Eminent Technology, Odyssey, Speakercraft, Kimber

Fax (909) 985-6968 Nordost Flatline, Synergistic, and much more